DSSM: Dynamic Streaming Pipeline With Spatiotemporal Model Toward Remote Sensing Object Detection

Shilong Jing, Yuchen Zhao, Zhuoxian Li, Xiangzhi Li, Xianda Xu, Hechong Wang, Hengyi Lv, Ming Sun, Yisa Zhang, and Yang Feng

***Abstract*—In the domain of visual perception tasks for uncrewed aerial vehicle (UAV) remote sensing, conventional object detection algorithms suffer severe challenges arising from drastic variations in perspective and complex backgrounds, proving particularly problematic for the identification of occluded and fast-motion targets. To address these issues, this article proposes a novel dynamic streaming architecture that encompasses the following three solution approaches: 1) we have established a revolutionary dual-stage streaming data pipeline (e.g., DSS Dataloader, DSS Sampler, and DSS Transform) to perceive target locations with more enriched spatiotemporal states, leveraging spatial pipeline to expedite rapid convergence on static representations, and applying temporal pipeline to facilitate fine-grained refinement by inject dynamic spatiotemporal details; 2) we have devised a novel selective memory module (SMM) that utilizes decoupled gating units to adaptively encode and store critical spatiotemporal features pertinent to the target from historical frame sequences while furnishing abundant contextual priors and endows the model with the capacity to capture long-range temporal dependencies; and 3) we have proposed a spatiotemporal loss function designed to constrain and guide the streaming model through state-smoothing constraint, motion-response constraint, and gating-conflict penalty, thereby improving the localization performance of the model. Moreover, we introduce a comprehensively revised version of the VisDrone-VID dataset prompted by the identification of numerous instances of disorganized bounding box annotations and erroneous labels within the original compilation. Finally, extensive experiments demonstrate that the dynamic streaming pipeline with spatiotemporal model (DSSM) outperforms all other state-of-the-art detection methods, achieving 14.4%, 77.6%, and 58.4% mAP@0.5:0.95 on the VisDrone-VID, EMRS-Frame, and EMRS-Event datasets. The updated VisDrone-VID dataset and our code are released at https://github.com/Jushl/DSSM**

***Index Terms*—Object detection, remote sensing, spatiotemporal loss, streaming data pipeline.**

I. Introduction

UNMANNED aerial vehicles (UAVs) have catalyzed transformative paradigm shifts across numerous sectors, including intelligent surveillance, precision agriculture, disaster response, and autonomous navigation, while object detection serves as the cornerstone for environmental perception and decision-making, representing the core technological underpinning for these applications [1], [2], [3], [4], [5], [6], [7], [8], [9], [10]. However, the direct transplantation of conventional object detection frameworks that were primarily engineered for terrestrial visual tasks, to an aerial perspective, results in a significant degradation of performance [11], [12], [13], [14]. This bottleneck predominantly stems from the inherent data characteristics of UAV platforms: extreme variations in viewpoint and scale, aggressive ego-motion and irregular flight trajectories, coupled with the complex and dynamic background topographies. Furthermore, traditional paradigms rely on the analysis of independent static frames and reveal their fundamental limitations when confronted with the challenges of intermittent object visibility caused by occlusions, and high-velocity target motion (Fig. 1 illustrates the challenges in remote sensing detection). As a result, a key question still remains: how to effectively model intrinsic continuity and dynamic consistency in the time dimension?

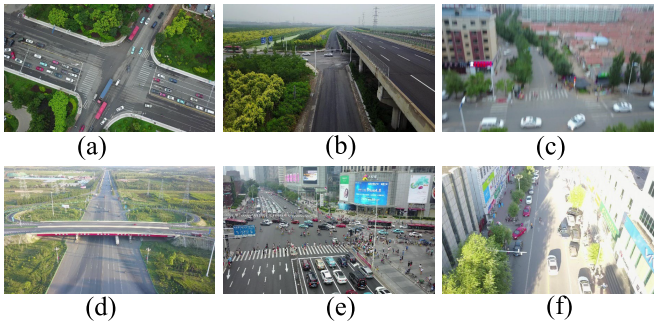


Fig. 1. Examples in challenging remote sensing scenario. (a) Normal, (b) background occlusion, (c) motion blur, (d) tiny target, (e) dense target, and (f) photometric interference.

To mitigate these challenges, a segment of existing research has shifted its focus toward the domain of streaming object detection, endeavoring to bolster the robustness of feature representations by exploiting the temporal correlations between historical frames [15], [16]. Recently, several approaches have explored leveraging diverse object properties across different time steps within temporal video streams to improve detection performance. Nevertheless, these methods are often contingent upon specific dataset sequences or highly correlated scenarios, exhibiting poor generalizability to disparate remote

Received 21 November 2025; revised 4 February 2026 and 26 March 2026; accepted 17 April 2026. Date of publication 22 April 2026; date of current version 1 June 2026. *(Corresponding author: Yuchen Zhao.)*

Shilong Jing, Zhuoxian Li, Xianda Xu, and Hechong Wang are with Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences, Changchun 130033, China, and also with the University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: jingshilong22@mails.ucas.ac.cn).

Yuchen Zhao, Xiangzhi Li, Hengyi Lv, Ming Sun, Yisa Zhang, and Yang Feng are with Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences, Changchun 130033, China (e-mail: zhaoyuchen@ciomp.ac.cn).

Digital Object Identifier 10.1109/TGRS.2026.3686603

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:38:48 UTC from IEEE Xplore. Restrictions apply.

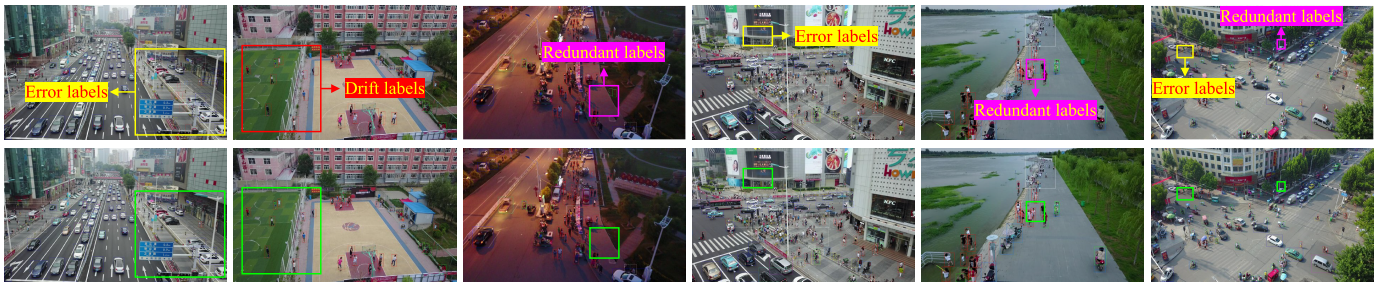


Fig. 2. Error and modified samples on the VisDrone-VID dataset. (First row) Original annotations and (second row) revised annotations, where the correct areas are marked in green, and the incorrect areas are marked in other colors.

sensing contexts [17], [18], [19]. Furthermore, other studies have investigated recurrent neural networks (RNNs) and 3-D convolutions to establish long-range temporal dependencies and enhance regression precision [20], [21]. While these techniques have improved detection stability, their suboptimal inference pipelines and exorbitant memory consumption preclude the efficient training of sophisticated spatiotemporal models. Importantly, these aforementioned approaches tend to homogeneous process spatiotemporal information, failing to draw a distinction between the static spatial representations and the dynamic temporal evolutions while resulting in suboptimal convergence during the initial training phases and impeding the effective capture of nuanced spatiotemporal cues. Moreover, concerning the modeling of long-term temporal dependencies, long short-term memory (LSTM) or Transformer networks are susceptible to information drift and the amnesia of critical features [22], [23], which is particularly pronounced when confronted with the background clutter and significant variations in target characteristics, thereby limiting the discriminative and persistent retention of target-relevant information.

To surmount the aforementioned limitations, this article aims to construct a groundbreaking dynamic streaming detection framework that employs spatial and temporal training stages, coupled with a selective memory module guided by a spatiotemporal loss to maintain the integrity of temporal information. Specifically, we establish a dual-stage streaming data pipeline comprising three components (e.g., DSS Dataloader, DSS Sampler, and DSS Transform), which initially utilizes a spatial data pipeline (SDP) to accelerate the progressive convergence of the model on static visual patterns within the frozen selective memory module (SMM). Subsequently, the SDP transitions to a temporal data pipeline (TDP) with unfrozen SMM and infuses dynamic information from the temporal dimension to achieve fine-grained model optimization. Here, the decoupled gating units of the SMM are engineered to adaptively encode and archive spatiotemporal features strongly correlated with the target from historical observation sequences, thereby providing rich contextual priors and establishing long-range temporal dependencies. Additionally, we propose a composite spatiotemporal loss function to impose effective constraints tailored for the streaming training architecture. By introducing stipulations for state smoothness, motion responsiveness, and a gating-conflict penalty as a high-order regularization term to guide the learning process of the model on complex spatiotemporal manifolds. Moreover, as a significant ancillary contribution, we have conducted a comprehensive review and rectification of the widely used VisDrone-VID dataset [24], correcting the substantial annotation noise and labeling errors. This effort provides the research community with a more reliable benchmark for algorithmic evaluation and comparison (Fig. 2 presents a comparative illustration on the VisDrone-VID dataset). Furthermore, comprehensive experimental results demonstrate that the proposed DSSM achieves 14.4%, 77.6%, and 58.4% mAP@0.5:0.95 on the rectified VisDrone-VID, EMRS-Frame, and EMRS-Event datasets, respectively, outperforming all state-of-the-art object detectors.

Our contributions can be summarized as follows.

1) We construct an innovative dual-stage streaming data pipeline that efficiently builds data batches from disparate video streams through a multithreaded parallelization mechanism. This pipeline concurrently mitigates the dependency on specific data sequences of the model and expedites training convergence.
2) We propose a novel selective memory module (SSM) that employs decoupled gating units to selectively forget and update historical states propagated through temporal dimension, preserving critical target details and achieving more stable sequential memory.
3) We design a composite spatiotemporal loss function by leveraging state smoothness, motion responsiveness, and gating penalty to guide and regulate the learning process of the SSM within complex spatiotemporal flows specifically for the streaming architecture, thereby enhancing the perception and location performance of the model.
4) We have revised the VisDrone-VID dataset while providing a comprehensively rectified version with original annotation methodology. This contribution introduces a reliable benchmark for the equitable comparison.
5) As a pioneering streaming object detector, DSSM achieves competitive performance compared to conventional detectors on both the vision-based VisDrone-VID dataset and the multimodal EMRS dataset.

II. Related Work

A. Streaming Architecture

Given that the limited spatial information afforded by individual frames, there are many studies have delved into the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:38:48 UTC from IEEE Xplore. Restrictions apply.

exploration of spatiotemporal data, aiming to leverage the intrinsic feature priors within video streams to surmount the bottlenecks encountered by conventional detection algorithms. ASTMNet [18] pioneered the use of an LSTM-like memory module to capture spatiotemporal features, introducing a novel paradigm for object detection. Subsequently, SODFormer [17] refined this methodology, employing a sequential sampler and Transformer framework for streaming inference to achieve better detection performance. Furthermore, ESVT [1] proposed a batch-parallelized sampling strategy to enhance the generalization capabilities of the model through a multipipeline training approach. Nevertheless, these methods are commonly constrained by the single-threaded data loader, which impedes model training and convergence. Additionally, RVT [20] and MVT [25] designed an alternative streaming training paradigm utilizing a multithreaded format, achieving notable results on event-based datasets. However, these strategies are hampered by a restricted memory window stemming from its cyclical data-loading mechanism, precluding the capture of spatiotemporal details over long-range dependencies. In conclusion, while the aforementioned methods have established a foundational basis for streaming architectures, they are still contending with the challenges of suboptimal convergence and constrained temporal memory. This obstacle arises from the necessity for the model to concurrently learn dynamic spatiotemporal transformations and precise spatial geometric features during the initial training stages, where the complex gradient landscape hinders the effective representation of deep semantic information and compromises the localization accuracy. Consequently, we introduce a novel streaming framework characterized by a dual-stage spatial and TDP. This framework is furnished with a complementary match of samplers, loaders, and data augmentation techniques, thereby providing a new perspective for challenging remote sensing detection.

B. Spatiotemporal Model

It is important to store memory and propagate spatiotemporal information. The LSTM network [22] has been widely applied to various sequential prediction tasks. Subsequently, the gated recurrent unit (GRU) [26] refines this storage mechanism, facilitating a more streamlined retention of details. Moreover, the Transformer attention [23] is used to capture long-range dependencies, while the progressive accumulation of historical tokens introduces a quadratic escalation in computational complexity. Furthermore, Mamba [27] employs a novel state space model to memorize spatiotemporal cues; however, its structured representation exhibits limitations in supporting the continuous and complex variations inherent in dynamic spatiotemporal streams. Therefore, drawing inspiration from the design principles of these aforementioned components, we propose an SMM with decoupled gates as trainable regulatory parameters, which is engineered to achieve robust long-distance memory in conjunction with a spatiotemporal loss function.

C. Object Detector

Currently, the mainstream methodologies in object detection can be broadly categorized into two families: 1) the single-stage detectors epitomized by the YOLO series [28] and 2) the end-to-end detection frameworks represented by DETR variants [29]. These two architectures collectively constitute the bedrock of remote sensing detection systems.

In the evolution of single-stage detectors, the YOLO framework has assumed a dominant position due to its exceptional tradeoff between computational efficiency and detection accuracy. The progression from YOLOv5 [30] that introduced cross-stage partial networks to enhance the representation of diminutive objects, to the refined label assignment strategies in YOLOv6 [31] and the anchor-free design coupled with a task-decoupled head in YOLOv8 [32], has progressively improved model generalization and robustness. More recent explorations, such as YOLOv10 [33], YOLOv11 [34], and YOLOv12 [35], have further investigated programmable gradients, lightweight network architectures, and the deep integration of deep convolution with attention mechanisms. Generally speaking, the common denominator among these approaches is their achievement of exceptional detection performance on static images through the optimization of network topologies and training strategies. It is worth noting that recent research has also begun to focus on how to extend the YOLO architecture across modalities for specific challenging environments (such as low-light conditions), where DNC-YOLO [36] designed a parallel skeleton to extract RGB and infrared features separately, and introduced distance-weighted IOU to improve localization accuracy. Meanwhile, a completely different paradigm emerged with the advent of DETR, which, through a core encoder–decoder and global attention mechanism, obviated the need for hand-engineered components like nonmaximum suppression. However, its substantial computational complexity and protracted convergence speed constituted its principal initial bottlenecks. To solve these challenges, subsequent research introduced various improvements: Deformable-DETR [37] incorporated a deformable attention mechanism, significantly mitigating the computational overhead on high-dimensional feature maps; DINO [38] proposed an innovative denoising training paradigm, achieving better detection performance; RT-DETR [39] and the variant RT-DETRv2 [40] streamlined the decoder architecture to substantially enhancing inference time; and D-FINE [41] further unlocked the performance potential of this end-to-end framework by redefining the bounding box regression mechanism.

Nevertheless, despite the remarkable achievements of these two architectural families, they share a fundamental limitation in their design philosophy; both are inherently engineered to process independent static image frames. Consequently, when directly applied to continuous video streams acquired from UAV platforms, this “frame-by-frame” processing paradigm fails to effectively exploit the inherent temporal coherence between frames. As a result, these models suffer from unstable detection performance when confronted with challenges precipitated by aggressive motion and background occlusions. Therefore, our work aims to outperform the static processing paradigm by proposing a novel detector capable of efficiently modeling spatiotemporal data streams, which is complemented by the designed streaming architecture to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:38:48 UTC from IEEE Xplore. Restrictions apply.

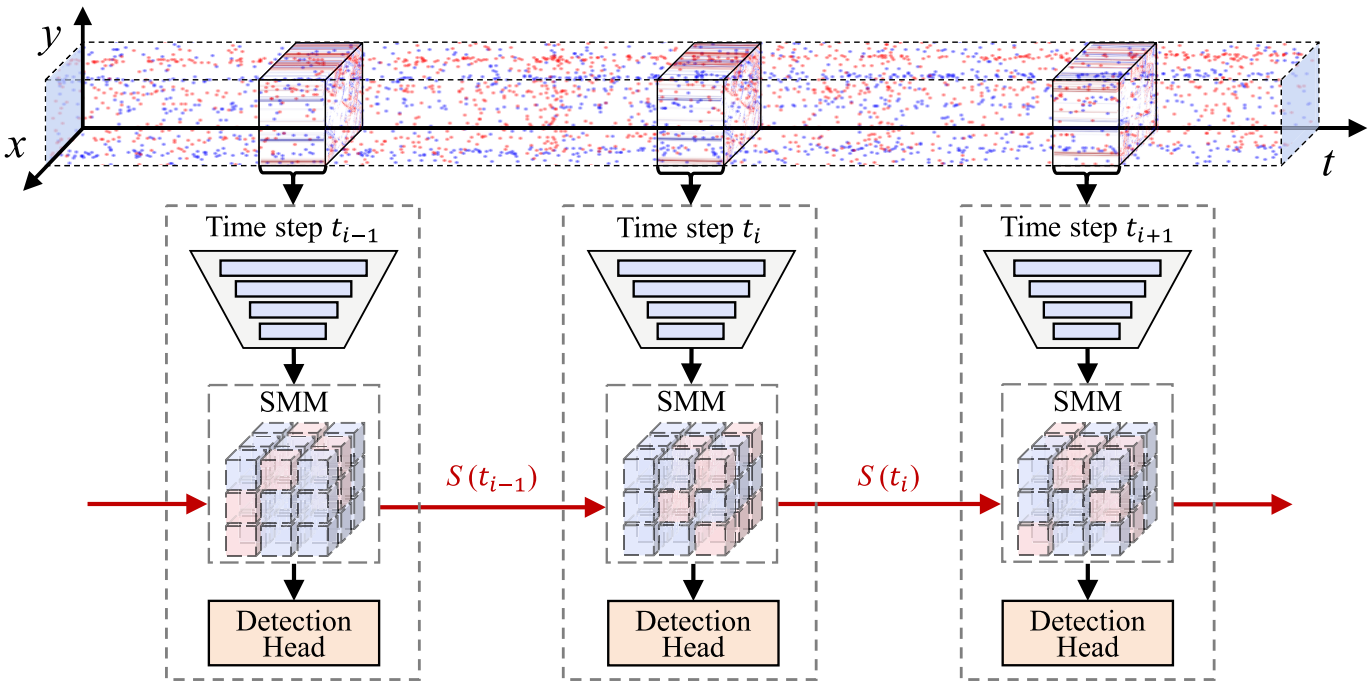


Fig. 3. Model overview of DSSM, which comprises four key components. (a) Data preprocessing that generates parallelized batches according to different input paradigms. (b) Backbone encoder that extracts feature maps with different scales. (c) SMM is used to memory historical cues and transport spatiotemporal cues. (d) Detection head receives the feature maps for fusion, which are subsequently fed into the decoupled head to predict categories and bounding boxes.

address the formidable challenges of remote sensing detection tasks.

III. Method

We present the overall framework of our model in Section III-A. Subsequently, the dual-stage streaming data pipeline is introduced in Section III-B. Additionally, we detail the selective memory module in Section III-C. Finally, we elucidate the spatiotemporal loss function in Section III-D.

A. Model Overview

As illustrated in Fig. 3, the overall model architecture is composed of four primary components: 1) data preprocessing is designed to generate parallelized data batches according to different input paradigms; 2) backbone encoder is constructed from the efficient modules that widely utilized in the YOLO series to extract feature semantic information; 3) selective memory module is tasked with propagating historical spatiotemporal states and outputting currently feature representation; and 4) feature pyramid network (FPN) and a decoupled detection head are employed to decode the feature maps and predict the classification and localization information, respectively.

1) Data Preprocessing: The input data are partitioned based on its attributes into visible and event-based modalities. For the visible frames, they are directly organized into a batch $Batch \in \mathbb{R}^{B\times C\times H\times W}$ through the streaming pipeline. For the event data, we utilize a voxel grid as the event representation to compose a corresponding batch $Batch \in \mathbb{R}^{B\times T\times H\times W}$, where the voxel grid generates frame-like tensors by applying a bilinear voting scheme within the selected temporal window at various event timesteps.

2) Backbone Encoder: We construct a cross-stage partial darknet (CSPDarkNet) by referencing mainstream YOLO architectures (e.g., YOLOv8 [32], YOLOv11 [34], and YOLOv12 [35]), thereby generating feature maps $f = [f_1, f_2, f_3, f_4]$ at various scales through four downsampling operations, which are subsequently utilized for fine-grained label classification and position regression. The entire process can be formulated as

$$f = \left[f_1, f_2, f_3, f_4\right] = Backbone\left(Batch\right). \tag{1}$$

3) Selective Memory Module: As the core component for spatiotemporal transformation, SMM is used to update the memory state matrix $M \in \mathbb{R}^{B\times C'\times H'\times W'}$ and yield an enhanced spatial representation $f' \in \mathbb{R}^{B\times C'\times H'\times W'}$. In this article, we apply this module exclusively to the deepest semantic feature $f_1 \in \mathbb{R}^{B\times C'\times H'\times W'}$, and this operation can be formulated as

$$\left[f_1'\left(t_i\right), state\left(t_i\right)\right] = SMM\left(\left[f_1\left(t_i\right), state\left(t_{i-1}\right)\right]\right). \tag{2}$$

4) Detection Head: Herein, only the first three-scale features $[f_1', f_2, f_3]$ are selectively processed by the BiFPN to achieve multilevel feature fusion. Subsequently, they are fed into a decoupled detection head to generate predictions for class $cls \in \mathbb{R}^{B\times L\times N}$ and bounding box $bbox \in \mathbb{R}^{B\times L\times 4}$, as depicted in the following equation:

$$[cls, bbox] = DH\left(BiFPN\left(\left[f_1', f_2, f_3\right]\right)\right). \tag{3}$$

B. Dual-Stage Streaming Data Pipeline

The dual-stage streaming data pipeline is composed of SDP and TDP. Each of these incorporates DSS Sampler, DSS dataloader, and DSS transform.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:38:48 UTC from IEEE Xplore. Restrictions apply.

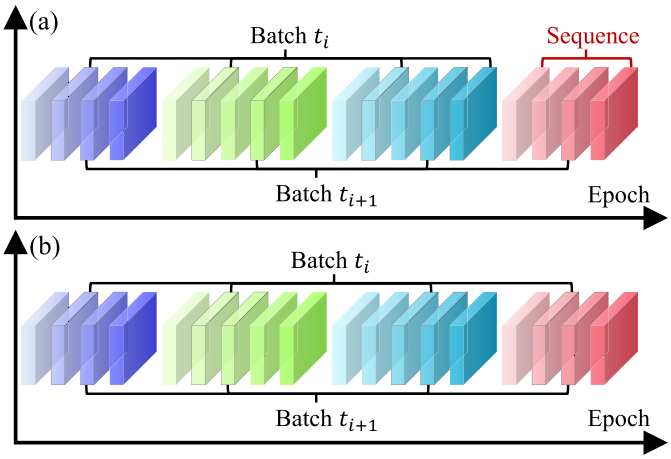


Fig. 4. Illustration of the DSS data pipeline computing architecture. (a) SDP and (b) TDP.

Fig. 4 illustrates the operational workflow of the DSS Data Pipeline. Within this framework, the SDP adheres to a random sampling strategy and incorporates complex geometric augmentations, whereas the TDP applies soft sequential transformations to enrich the training data. Specifically, each batch sampled by the SDP is randomized, and its enhancement methods (e.g., mosaic splicing, random flip, and photometric distortion) are generated stochastically during the forward pass. In contrast, the TDP employs a streaming flip mechanism to determine the augmentation strategy before each iteration while loading new video sequences when a sequence group within the batch becomes depleted. This approach thereby facilitates multithreaded data ingestion while ensuring continuous, parallelized data reading.

Moreover, the proposed DSS data pipeline exclusively loads the current frame position of each video sequence during sampling and randomly selects a new sequence within the batch when a given sequence is exhausted. Meanwhile, the memory state is propagated along the temporal dimension across batches while being detached from the computation graph, thereby supporting gradient updates without causing information leakage. Generally, this mechanism adheres to the rigorous criteria of real-time streaming processing (perform inference based solely on historical frames without seeking future information), accommodating both multisequence parallel learning during training and frame-by-step processing during inference.

1) DSS Sampler: The DSS Sampler aims to address the challenge of efficiently constructing batches for processing streaming temporal video data, employing a dynamic resource allocation mechanism to achieve a balanced loading of multisource data while preserving the integrity of each batch compared to the conventional sampling strategies.

Specifically, given a dataset $\mathcal{D}$ containing N independent video sequences, where each sequence V_i comprises n_i image frames, we construct a global index mapping function $\Phi : (i, j) \rightarrow k$, where i denotes the video index, j represents the frame index, and k serves as the globally unique identifier. The data structure can be formally expressed as

$$\mathcal{S} = \{(s_i, e_i, n_i)\}_{i=1}^{N}. \tag{4}$$

Here, $s_i = \sum_{m=1}^{i-1} n_m$ denotes the starting index and $e_i = (s_i + n_i - 1)$ represents the ending index. Furthermore, during the initialization phase, a video metadata table $\mathcal{M}$ is established to log the global index range for each video. Assuming a batch size of B, we initialize a current read position vector $\mathbf{p} = [0]^N$ and an active video set $\mathcal{A} \subseteq \{1, \ldots, N\}$, where the cardinality of this set is defined as $|\mathcal{A}| = \min(B, N)$. For each active video $v \in \mathcal{A}$, the frame $\Phi(v, p_v)$ is sampled and incremented p_v if $p_v < n_v$. Meanwhile, a replacement strategy is executed when a video is exhausted

$$replace\,(v) = argmin_{u \notin \mathcal{A}} \{disk\,(u, b)\,|p_u < n_u\} \tag{5}$$

where the distance function $dist(u, v)$ preferentially selects available videos that have adjacent indices. Moreover, an expansion algorithm is performed when the batch is not full

$$\mathcal{A}' = \mathcal{A} \cup argmax_{u \notin \mathcal{A}} \{n_u | p_u < n_u\}. \tag{6}$$

Equation (6) is applied repeatedly until either the batch is full or no more supplementary videos are available. By preserving batch diversity while precluding the unproductive sampling of residual frames from any single video, the DSS Sampler improves the training efficacy on extended video sequence data, thereby furnishing a data sampler basis for sequential modeling tasks.

2) DSS Dataloader: To address the requirements for spatiotemporal feature learning in the context of streaming object detection, DSS Dataloader utilizes a dynamic adaptive strategy to intelligently decouple the training process into a spatial feature extraction phase and a temporal relationship modeling phase, thereby enabling a temporally aware transition.

First, we establish a hierarchical data structure for the management of video sequences, organizing them on a per-scene basis, where the central principle is to characterize each scene as a discrete data unit

$$\mathcal{D}_i = \left\{\left(I_j, A_j\right)\right\}_{j=1}^{N_i} \tag{7}$$

where I_j represents an image frame and A_j denotes its corresponding annotation file, which contains the bounding box and label information. During the initialization phase, all scene units are randomly shuffled to enhance the generalization capability of the network, while meticulously preserving the scene temporal continuity among frames. In addition, the data access employs an efficient coordinate mapping mechanism generated by the sampler, which translates a global index into a scene-local index pair to facilitate random access while retaining critical temporal contextual information.

Moreover, during the SDP stage, we utilize a standard random sampler to compel the model to concentrate on learning robust spatial appearance features and object recognition capabilities. Conversely, in the TDP stage, DSS dataloader automatically transitions to a temporal mode using the DSS Sampler, ensuring that each training batch is composed of a continuous sequence of frames originating from the same scene. Furthermore, to resolve the issue of matching sequence lengths with the batch size, we introduce a sequence padding mechanism by randomly duplicating existing elements within a sequence, thereby providing a coherent input for the temporal framework.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:38:48 UTC from IEEE Xplore. Restrictions apply.

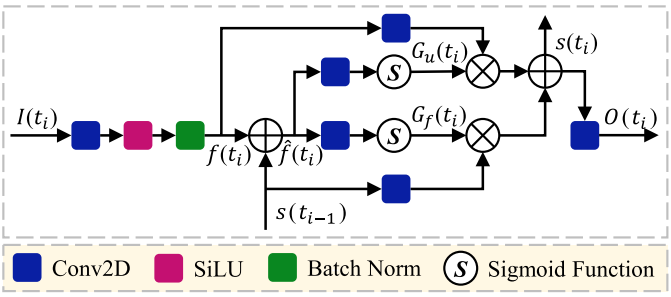


Fig. 5. Illustration of the SMM, which establishes a bridge for the integration of information across the spatial and temporal domains.

3) DSS Transform: To accommodate the distinct characteristics of the training and inference phases within object detection, the DSS transform is engineered for the dynamic adaptation of data augmentation. This paradigm applies a mode-aware strategy of transformation combinations, introducing geometric and photometric perturbations during the training stage to improve model robustness, while adopting deterministic transformations during the inference stage to guarantee evaluation consistency.

Specifically, DSS Transform is delineated into two core stages: the first stage leverages a rich set of geometric augmentations and color distortions to enhance the visual contextual information, compelling the model to learn object features across diverse local environments; the second stage optimizes data-loading efficiency via a streaming processing mechanism, which forgoes computationally intensive augmentations (e.g., mosaic and visual perspective) instead employing streaming horizontal and vertical flips to maintain the generalization capability of the model.

This design, by decoupling the transformation logic for the training and inference stages, achieves a precise alignment between the data augmentation strategy and the task objectives, thereby enhancing model generalization and ensuring the reliability of the evaluation metrics.

C. Selective Memory Module

We introduce SMM based on the selective memory mechanism, a novel spatiotemporal feature propagation unit designed explicitly for sequential visual data. Its core innovation lies in a decoupled gating policy, which fundamentally differs from the coupled gating in classical RNNs (e.g., LSTM and GRU) by enabling independent control over the inflow of current information and the retention of historical memory while allowing SMM to adaptively model interdependencies across both temporal and spatial dimensions with a fully convolutional architecture.

As illustrated in Fig. 5, we first employ a convolutional encoder to capture the spatial context of input features and then regulate information flow via the synergistic yet independent operation of an update gate and a forget gate. Given the input feature tensor $I(t_i)$ at timestep t_i and the historical state $s(t_{i-1})$ (initialized as a zero tensor), the process can be unfolded as follows:

Local Spatial Enhancement: The current input $I(t_i)$ undergoes a nonlinear transformation to enhance its local structural representation

$$f(t_i) = Norm(SiLU(Conv2D(I(t_i)))) \tag{8}$$

where the convolution uses a 3×3 kernel with padding 1. This step provides a spatially aware foundational feature, distinguishing our approach from the direct linear projection often used in standard RNNs.

Cross Feature Priming: An additive fusion establishes initial correlation between the current observation $f(t_i)$ and the last state $s(t_{i-1})$, forming a basis for subsequent gating decisions

$$\hat{f}(t_i) = f(t_i) + s(t_{i-1}). \tag{9}$$

Decoupled Gating Signal Generation: Critically differing from a single gate derived from concatenated features of GRU and LSTM, SMM employs parallel and independent convolutional pathways to generate two separate gating signals from the enhanced feature $f(t_i)$

$$\begin{aligned} G_u(t_i) &= \sigma(Conv2D(f(t_i))) \\ G_f(t_i) &= \sigma(Conv2D(f(t_i))). \end{aligned} \tag{10}$$

Here, σ is the sigmoid function, and each convolution uses a 1×1 kernel for channel-wise linear transformation. This decoupling strategy allows G_u and G_f to make independent decisions, respectively.

Spatially Aware Memory Update: The state update further incorporates spatial modeling via depthwise separable convolutions before applying the gating weights

$$\begin{aligned} f'(t_i) &= Conv2D(f(t_i)) \\ s'(t_{i-1}) &= Conv2D(s(t_{i-1})) \\ s(t_i) &= G_u(t_i) \odot f'(t_i) + G_f(t_i) \odot s'(t_{i-1}) \end{aligned} \tag{11}$$

where $\odot$ denotes element-wise multiplication and the convolutional operation with a depthwise separable 3×3 kernel, which achieves sparse interchannel parameter interaction through the grouped convolutions. Generally speaking, this update rule synthesizes the gating principles inherent to both LSTM and GRU while also enabling context-aware memory modulation across the feature map.

Feature Projection: Finally, a 1×1 convolution projects the updated spatiotemporal state $s(t_i)$ back to a standard 2-D feature space

$$O(t_i) = Conv2D(s(t_i)). \tag{12}$$

In conclusion, SMM provides a theoretically grounded and interpretable framework for learning long-term dependencies in visual sequences, where the decoupled convolutional gating and spatially sensitive memory update offer a distinct architectural advantage over classical RNNs, facilitating more flexible and robust dynamic feature integration for tasks such as video object detection.

D. Spatiotemporal Loss

To improve the stability and discrimination of temporal features in streaming video comprehension, we propose a joint optimization loss that integrates spatiotemporal consistency

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:38:48 UTC from IEEE Xplore. Restrictions apply.

constituted by three core components (State smoothness regularization, motion-aware update constraint, and gate exclusion penalty) to impose structured constraints on the dynamics of the hidden state and the gating mechanisms, regularizing the update policy of the memory unit explicitly.

1) State Smoothness Regularization: We introduce smoothness regularization to penalize unnecessary fluctuations of the hidden state in the temporal dimension. Simply, this limitation is predicated on adaptively weighting the state differential by the gating signals and is specifically defined as

$$\mathcal{L}_{smooth} = \|(s(t_i) - s(t_{i-1})) \odot (1 - G_u(t_i))\|_2^2 \tag{13}$$

where $s(t_i)$ denotes the spatiotemporal state at timestep t_i and $G_u(t_i)$ represents the update gate vector, which provides a gate-aware weighting on the state variation through $(1 - G_u(t_i))$. Particularly, when the model exhibits a tendency to preserve historical information (i.e., as $G_u(t_i)$ approaches 0), this mechanism reinforces state continuity and suppresses feature oscillations arising from ineffectual updates, thereby enhancing the robustness of the temporal modeling.

2) Motion-Aware Update Constraint: We design motion-driven consistency to incentivize the gating mechanism to respond in concert with the dynamic variations within the scene. Correspondingly, by aligning the update gate with the magnitude of the feature differential, this loss guides the model to adaptively modulate the intensity of information updates based on the degree of motion, which can be formally expressed as

$$\mathcal{L}_{motion} = \|G_u(t_i) - \sigma(|f(t_i) - f(t_{i-1})|)\|_2^2 \tag{14}$$

where $f(t_i)$ denotes the features of the current frame and σ represents the sigmoid activation function. Here, the magnitude of the feature difference serves as a proxy signal for scene dynamics, which may arise from object motion, illumination variation, or background shift. Nevertheless, the model is guided to learn to focus on salient changes that are consistent across space and time under the joint effect of state smoothness regularization, thereby suppressing the influence of transient disturbances in an end-to-end optimization framework. This constraint encourages the gating vector to correlate positively with the magnitude of interframe feature variation. As a result, it prioritizes memory updates in regions with significant motion while retaining historical information in static areas, mimicking the attentional mechanism of the human visual system.

3) Gate Exclusion Penalty: To preclude semantic conflicts that arise from the simultaneous activation of the forget and update gates, we apply a penalty term for their overlap during the training process

$$\mathcal{L}_{gate} = |G_f(t_i) \odot G_u(t_i)|_1. \tag{15}$$

By minimizing the element-wise product of the gate vectors, we encourage the formation of a complementary distribution between them. Equation (15) improves the distinctness of the spatial responsibilities of the gating units, enhancing both the interpretability and the efficiency of the memory update mechanism.

TABLE I
REVISION DETAILS OF THE VISDRONE-VID DATASET. THIS CONTRIBUTION INTRODUCES A RELIABLE BENCHMARK FOR THE EQUITABLE COMPARISON

Dataset	Image	Annotation	Revised Image	Revised Annotation
Train	24,198	1,158,853	3,763	21,602
Test	2,846	116,713	597	2,852
Total	27,044	1,275,566	4,360	24,454

In summary, the total optimization objective is the weighted sum of these three structured losses

$$\mathcal{L}_{st} = \lambda_1 \mathcal{L}_{smooth} + \lambda_2 \mathcal{L}_{motion} + \lambda_3 \mathcal{L}_{gate} \tag{16}$$

where $\lambda_1 = 1.0$, $\lambda_2 = 0.5$, and $\lambda_3 = 0.2$ serve as the balancing hyperparameters that will be discussed in Section IV-C4. Through the joint optimization of the aforementioned losses, DSSM is conditioned to learn streaming representations in a manner that is temporally coherent, motion-adaptive, and gate-discriminative, thereby significantly elevating inference performance in complex dynamic scenarios.

IV. EXPERIMENTS

In Section IV-A, we introduce the revised VisDrone-VID dataset [24] and the EMRS dataset [1]. Subsequently, in Section IV-B, we elaborate the implementation details (e.g., model variants, training settings, and evaluation metrics). Furthermore, Section IV-C provides ablation studies for the individual components. Finally, Section IV-D presents a benchmark comparison between the proposed method and other state-of-the-art approaches.

A. Datasets

1) VisDrone-VID Dataset: This consists of 27 044 images, allocated into 24 198 for training and 2846 for testing, while encompassing ten distinct object categories: pedestrian, people, bicycle, car, van, truck, tricycle, awning-tricycle, bus, and motorcycle. Furthermore, we correct mislabeled instances in the VisDrone-VID dataset as illustrated in Table I, involving 3763 images from the training set and 597 from the test set, amounting to a total of 24 454 revised annotations (e.g., cluttered background, wrong category, redundant annotations, bounding box drift, and missing annotations).

2) EMRS Dataset: This contains 10 800 asynchronous data pairs (frames and event streams) captured across various scenarios (e.g., normal, motion blur, and low-light conditions), including 9500 samples for training and 1300 for validation, while encompassing five common object categories: car, bus, truck, pedestrian, and two-wheel.

B. Implementation Details

We begin in Section IV-B1 by introducing three model variants designed to analyze performance across different network depths. Subsequently, Section IV-B2 provides a detailed exposition of the parameter configurations and training methodology. Finally, the critical evaluation metrics

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:38:48 UTC from IEEE Xplore. Restrictions apply.

TABLE II
DETAILS OF THE MODEL VARIANTS. ALL PARAMETERS EXCEPT THOSE EXPLICITLY SPECIFIED REMAINED CONSISTENT THROUGHOUT THE EXPERIMENTS

Variant	Backbone	Channels		
		Scale1	Scale2	Scale3
DSSM-N	CSPDarkNet 26	64	128	256
DSSM-S	CSPDarkNet 26	128	256	512
DSSM-B	CSPDarkNet 36	192	384	768

(e.g., mean average precision, model parameters, and computational complexity) are delineated in Section IV-B3.

1) Model Variants: To assess the scalability of our proposed model, we build three distinct variants by modulating the number of convolutional blocks in the backbone architecture and adjusting the dimensionalities of the three extracted feature maps. As summarized in Table II, the Nano and Small variants share an identical encoder structure comprising 26 convolutional blocks, yet differ in their feature scales: the Nano variant extracts feature maps with channel dimensions of 64, 128, and 256, whereas the Small variant produces features with 128, 256, and 512 channels, respectively. Furthermore, the Base variant incorporates a CSPDarkNet backbone consisting of 36 convolutional blocks, generating feature maps at scales of 192, 384, and 768 channels.

2) Training Settings: In this work, the following training protocol is adopted: the batch size is set to 64, the number of data-loading workers to 16, the initial learning rate to 0.01, and the weighting coefficients for the individual components of the loss function are configured as follows: 7.5 for bounding box regression loss, 0.5 for classification loss, and 1.5 for distribution focal loss (DFL). In addition, the AdamW optimizer is employed with a weight decay coefficient of 5×10^{-4}, and all experiments are conducted with a fixed random seed of 42 to ensure reproducibility. Moreover, automatic mixed precision (AMP) training is incorporated to enhance computational efficiency and throughput. Furthermore, the training procedure spanned 72 epochs, divided into two key phases: the first 64 epochs constituted the SDP, focusing on spatial structure modeling, while the subsequent 8 epochs formed the TDP, aimed at strengthening the capacity of the model for capturing temporal details. Specifically, the selective memory module is kept frozen to prevent potential confusion in recursive state information caused by mosaic augmentation during the SDP, while in the TDP, the corresponding module is unfrozen to learn temporal dynamic features.

As illustrated in Fig. 6, we incorporate a variety of data augmentation strategies, including horizontal flipping, vertical flipping, photometric distortion, mosaic composition, and perspective transformation. Specifically, the SDP integrates mosaic composition, perspective transformation, random photometric distortion, random horizontal flipping, and random vertical flipping. In contrast, the TDP encompasses random photometric distortion, streaming horizontal flipping, and streaming vertical flipping. It is noteworthy that the key distinction in flip augmentation between SDP and TDP lies in their decision mechanisms: SDP dynamically determines whether to apply flipping concurrently with other image transformation operations, whereas TDP predefines the flipping strategy prior to the data augmentation pipeline, ensuring a balance between spatial consistency and temporal continuity in spatiotemporal modeling.

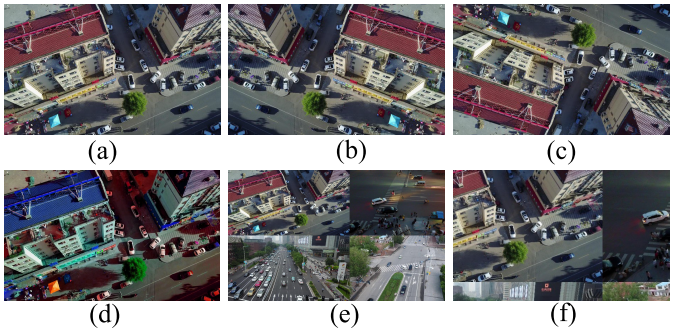


Fig. 6. Illustration of various augmentation methods. The SDP incorporates mosaic composition and perspective transformation, whereas the TDP employs a streaming strategy to preserve interframe temporal consistency. (a) Raw frame. (b) Horizontal flip. (c) Vertical flip. (d) Photometric distort. (e) Mosaic. (f) Perspective.

3) Evaluation Metrics: Within the evaluation paradigm of object detection, precision (P) and recall (R) constitute the fundamental dimensions for performance quantification. Precision is defined as the ratio of true positives (TPs) to all positive predictions [i.e., the sum of TP and false positives (FPs)], thereby characterizing the specificity in avoiding false alarms. Meanwhile, recall is expressed as the proportion of TP to all actual positive instances [i.e., the sum of TP and false negatives (FNs)], reflecting the capacity of the model to cover positive samples. These metrics are formally defined as

$$P = \frac{TP}{TP + FP}$$
$$R = \frac{TP}{TP + FN}. \tag{17}$$

Based on this foundation, a precision–recall (P–R) curve can be yielded by systematically varying the detection confidence threshold, which characterizes the tradeoff between precision and recall across different operational points. As a result, the average precision (AP) metric quantifies this relationship by computing the area under the P–R curve. Specifically, an interpolation method is applied to approximate the integral, thereby mitigating potential bias induced by the nonmonotonic nature of the curve. This process can be formulated as

$$AP = \int_0^1 P(R)\, dR. \tag{18}$$

Subsequently, the mean AP (mAP) extends the AP concept by computing the arithmetic mean of AP values across all categories, providing a holistic evaluation of multiclass detection performance. This metric can be formally expressed as

$$mAP = \frac{\sum_{i=1}^{C} (AP_i)}{C}. \tag{19}$$

Furthermore, averaging performance across multiple intersection over union (IoU) thresholds promotes a more equitable evaluation of model robustness. For instance, mAP@0.5

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:38:48 UTC from IEEE Xplore. Restrictions apply.

TABLE III

ABLATION OF SMM IN THE EMRS-FRAME DATASET. THE BEST DETECTION PERFORMANCE IN BOLD

Model	params (M)	mAP @0.5:0.95	mAP @0.5	mAP @0.75
Baseline	11.14	0.715	0.913	0.810
LSTM	**10.48**	0.734	0.936	0.827
SMM (ours)	10.74	**0.768**	**0.950**	**0.851**

denotes the mAP calculated at a single IoU threshold of 0.5, emphasizing detection performance under moderate overlap criteria. Meanwhile, mAP@0.75 employs a more stringent IoU threshold of 0.75, focusing on scenarios demanding superior localization accuracy. As for mAP@0.5:0.95, which is computed by averaging AP values across IoU thresholds ranging from 0.5 to 0.95 with a step size of 0.05, thereby achieving a comprehensive performance indicator that encompasses localization requirements from lenient to rigorous. These metrics are generally regarded as a pivotal benchmark for assessing model generalization capability.

In addition to detection accuracy metrics, the evaluation of computational efficiency is equally crucial. GFLOPs (giga floating-point operations) quantify the number of floating-point operations required for a single forward pass of the model, thus serving as an indicator of computational complexity and hardware resource demands. Moreover, the parameter count reflects the capacity of the model and is typically associated with memory footprint. Meanwhile, frames per second (FPS) measures the inference speed, representing an important performance indicator for real-time applications. These metrics collectively form a multidimensional evaluation framework that ensures a comprehensive analysis between accuracy and efficiency, aligning with the practical requirements of modern object detection systems in edge computing and large-scale deployment scenarios.

C. Ablation Experiments

To ensure impartial analysis of the designed components, we maintain identical model parameters and training configurations across a consistent experimental platform. Specifically, we utilize the DSSM-S model for evaluations on the EMRS-Frame dataset.

1) Ablation of the SMM: It is a novel strategy that leverages spatiotemporal historical cues for object detection, aiming to enhance localization performance by modeling inter-frame temporal relationships, where the specifically designed SMM component is grounded in this fundamental principle. Afterward, we conduct a progressive ablation study: initially establishing a performance baseline with the spatial model, subsequently incorporating an LSTM module to validate potential benefits of temporal modeling, and ultimately replacing it with our proposed SMM module to selectively improve pivotal frame features while suppressing extraneous information across the temporal dimension.

As shown in Table III, the spatial-based baseline model achieves 71.5% mAP@0.5:0.95, 91.3% mAP@0.5, and 81.0% mAP@0.75. Moreover, the LSTM with temporal modeling improves mAP@0.5:0.95 by 1.9%, reaching 73.4% mAP@0.5:0.95, 93.6% mAP@0.5, and 82.7% mAP@0.75. Furthermore, the more advanced SMM outperforms the baseline by 5.3% mAP@0.5:0.95, achieving 76.8% mAP@0.5:0.95, 95.0% mAP@0.5, and 85.1% mAP@0.75. Correspondingly, Fig. 7 demonstrates that the introduction of LSTM and SMM modules brings better attention characteristics compared to the baseline configuration.

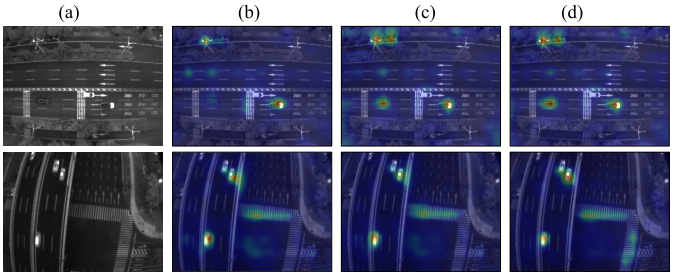


Fig. 7. Comparisons of the heatmap. (a) Raw frame. (b) Baseline model devoid of any spatiotemporal components. (c) Baseline with LSTM. (d) Baseline with SMM.

TABLE IV

ABLATION OF GATED DECOUPLING IN THE EMRS DATASET. THE BEST DETECTION PERFORMANCE IN BOLD

Model	params (M)	mAP @0.5:0.95	mAP @0.5	mAP @0.75
Coupled strategy	13.24	0.757	0.936	0.834
Decoupled strategy	**10.74**	**0.768**	**0.950**	**0.851**

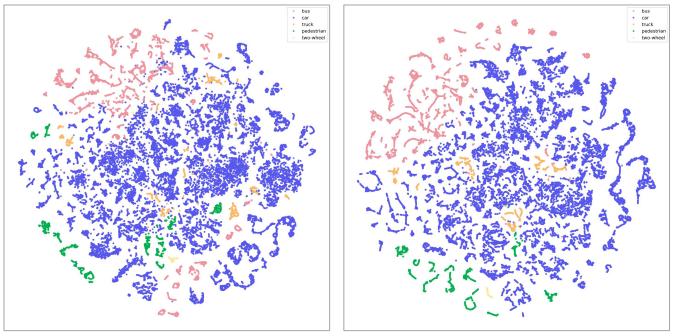


Fig. 8. tSNE of the gate response mechanism. (a) Coupled approach and (b) decoupled strategy.

2) Ablation of the Gated Decoupling: Conventional coupled gating mechanisms apply the joint generation approach to simultaneously produce the forget and update gates. However, this architecture suffers from inherent feature confounding issues; all gating signals share an identical feature transformation space, resulting in coupled decision boundaries that consequently undermine the selective memory capacity of the model. As evidenced in Table IV, the coupled strategy achieves 75.7% mAP@0.5:0.95, 93.6% mAP@0.5, and 83.4% mAP@0.75, while the decoupled strategy achieves consistent improvement across all metrics, achieving 76.8% mAP@0.5:0.95, 95.0% mAP@0.5, and 85.1% mAP@0.75.

Furthermore, Fig. 8 reveals the gating response analysis, where the decoupled strategy exhibits more discriminative

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:38:48 UTC from IEEE Xplore. Restrictions apply.

TABLE V

ABLATION OF DETECTION LOSS IN THE EMRS DATASET. THE BEST DETECTION PERFORMANCE IN BOLD

Model	Spatiotemporal Loss			mAP	mAP	mAP
	SS Loss	MR Loss	GP Loss	0.5:0.95	@0.5	@0.75
DSSM				0.739	0.929	0.810
	✓			0.757	0.946	0.842
		✓		0.751	0.947	0.833
			✓	0.743	0.930	0.817
	✓	✓	✓	**0.768**	**0.950**	**0.851**

clustering compared to the coupled strategy. This observation further validates the efficacy of the decoupling mechanism in modeling temporal consistency within complex scenarios.

3) Ablation of the Detection Loss: To systematically investigate the regulatory mechanism of the proposed spatiotemporal loss on structured state space models in dynamic visual perception tasks, we progressively introduce the state smoothness (SS), motion responsiveness (MR), and gating penalty (GP) loss to examine their synergistic effects on the learning process within complex spatiotemporal streams.

As shown in Table V, the baseline configuration without spatiotemporal loss achieves 73.9% mAP@0.5:0.95, 92.9% mAP@0.5, and 81.0% mAP@0.75; the incorporation of state smoothness loss achieves 75.7% mAP@0.5:0.95, 94.6% mAP@0.5, and 84.2% mAP@0.75; the incorporation of motion responsiveness loss results in 75.1% mAP@0.5:0.95, 94.7% mAP@0.5, and 83.3% mAP@0.75; and the incorporation gating penalty loss achieves 74.3% mAP@0.5:0.95, 93.0% mAP@0.5, and 81.7% mAP@0.75. Notably, the combined integration of all three loss components yields the best detection performance, achieving 76.8% mAP@0.5:0.95, 95.0% mAP@0.5, and 85.1% mAP@0.75.

4) Ablation of the Spatiotemporal Loss: We conduct an ablation study on the three components of the spatiotemporal loss to quantify their respective contributions under complex dynamic scenarios. Specifically, the state smoothness loss prevents undesirable data drift when the network should preserve memory, while the motion responsiveness loss emulates human visual attention by prioritizing dynamic regions, and the gating penalty loss improves the discriminative ability of the model.

As shown in Table VI, employing all three loss components with equal weighting achieves 74.9% mAP@0.5:0.95, 93.1% mAP@0.5, and 82.8% mAP@0.75 on the EMRS-Frame dataset, 54.3% mAP@0.5:0.95, 71.7% mAP@0.5, and 57.9% mAP@0.75 on the EMRS-Event dataset, and 12.1% mAP@0.5:0.95, 24.6% mAP@0.5, and 10.6% mAP@0.75 on the VisDrone-VID dataset. In contrast, reducing the influence of motion responsiveness and gating penalty terms by assigning respective weights of 1.0, 0.8, and 0.5, we achieve 76.0% mAP@0.5:0.95, 94.2% mAP@0.5, and 83.7% mAP@0.75 on the EMRS-Frame dataset; 54.7% mAP@0.5:0.95, 72.8% mAP@0.5, and 58.0% mAP@0.75 on the EMRS-Event dataset; and 12.8% mAP@0.5:0.95, 25.8% mAP@0.5, and 11.0% mAP@0.75 on the VisDrone-VID dataset. These observations align with our hypothesis that the state smoothness loss plays a predominant role in maintaining the stability

TABLE VI

ABLATION OF SPATIOTEMPORAL LOSS IN THE EMRS DATASET. THE BEST DETECTION PERFORMANCE IN BOLD

Dataset	Spatiotemporal Loss			mAP	mAP	mAP
	SS Loss	MR Loss	GP Loss	0.5:0.95	@0.5	@0.75
Frame	1.0	1.0	1.0	0.749	0.931	0.828
	1.0	0.8	0.5	0.760	0.942	0.837
	1.0	0.5	0.2	0.766	0.947	0.845
	1.0	0.5	0.1	**0.768**	**0.950**	**0.851**
Event	1.0	1.0	1.0	0.543	0.717	0.579
	1.0	0.8	0.5	0.547	0.728	0.580
	1.0	0.5	0.2	0.550	0.732	0.586
	1.0	0.5	0.1	**0.551**	**0.733**	**0.587**
VisDrone	1.0	1.0	1.0	0.121	0.246	0.106
	1.0	0.8	0.5	0.128	0.258	0.110
	1.0	0.5	0.2	0.132	**0.267**	0.118
	1.0	0.5	0.1	**0.133**	**0.267**	**0.119**

TABLE VII

ABLATION OF SDP IN THE EMRS DATASET, WHERE PERSPECTIVE TRANSFORMATION ARE COUPLED WITH MOSAIC SPLICING FOR DATA AUGMENTATION, AND THE BEST RESULT IN BOLD

H-Flip	V-Flip	Mosaic	mAP @0.5:0.95	mAP @0.5	mAP @0.75
			0.739	0.927	0.816
✓			0.744	0.931	0.828
	✓		0.742	0.925	0.86
		✓	0.751	0.948	0.842
✓	✓	✓	**0.768**	**0.950**	**0.851**

of historical memory, while the gating penalty term acts as a mild regularizer to mitigate conflicting behaviors during simultaneous forgetting and updating operations. As a result, DSSM achieves the best performance with hyperparameters set to 1.0 for SS loss, 0.5 for MR loss, and 0.1 for GP loss.

5) Ablation of the Transform Augmentation: Data augmentation mechanisms improve the robustness and generalization capability of the model by introducing perturbed samples into the training pipeline. In this work, the proposed SDP and TDP methodologies incorporate deeply structured composite transformation paradigms; thus, we conduct ablation studies to systematically evaluate their respective contributions.

Table VII presents the ablation analysis of individual components within SDP. Experimental results demonstrate that applying random horizontal flipping and vertical flipping independently achieves 74.4% and 74.2% mAP@0.5:0.95, respectively. Moreover, introducing a mosaic composition strategy further elevates detection performance to 75.1%, while the combination of all three augmentation techniques achieves 76.8% mAP@0.5:0.95. Notably, perspective transformation effectively enables diversified modeling of spatial geometric structures when integrated with mosaic image synthesis, thereby substantially enhancing data diversity.

Table VIII summarizes the ablation results for TDP components, where streaming horizontal flipping and vertical flipping achieve 76.0% and 75.9% mAP@0.5:0.95, respectively, while photometric distortion achieves 75.6% mAP@0.5:0.95. Furthermore, the simultaneous application of all three temporal augmentation methods achieves 76.8%

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:38:48 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
ABLATION OF TDP IN THE EMRS DATASET, WHERE THE CONSISTENT STRATEGY OF HORIZONTAL AND VERTICAL FLIPPING IS MAINTAINED THROUGHOUT THE SAME VIDEO STREAM, AND THE BEST RESULT IN BOLD

H-Flip	V-Flip	P-Distort	mAP @0.5:0.95	mAP @0.5	mAP @0.75
			0.754	0.937	0.835
✓			0.760	0.940	0.842
	✓		0.759	0.940	0.841
		✓	0.756	0.938	0.838
✓	✓	✓	**0.768**	**0.950**	**0.851**

TABLE IX
ABLATION OF TRAINING EPOCH IN THE EMRS DATASET. THE BEST DETECTION PERFORMANCE IN BOLD

Dataset	DSS Training Epoch			mAP	mAP	mAP
	Total	SDP	TDP	0.5:0.95	@0.5	@0.75
EMRS	72	72	0	0.749	0.931	0.828
	72	68	4	0.763	0.946	0.838
	72	64	8	**0.768**	**0.950**	**0.851**
	72	60	12	0.764	0.947	0.840
	72	0	72	0.727	0.926	0.811

mAP@0.5:0.95, demonstrating complementary effects in feature perturbation across temporal dimensions.

Collectively, these results indicate that SDP generates more substantial accuracy gains than TDP, attributable to its capacity to induce sufficient feature space perturbation through highly nonlinear transformations of input spatial structures. However, as training progresses, relying exclusively on SDP gradually leads to performance saturation in spatial bounding box regression tasks, manifested as reduced sensitivity to geometric deformations. To address this limitation, we propose a temporal perturbation mechanism as a complementary strategy, integrating spatiotemporal cues to synergistically optimize the perceptual domain of the model. This design rationale not only provides an innovative solution to mitigate inherent constraints of spatial augmentation but also constitutes the fundamental motivation for adopting the streaming training paradigm to continuously improve detection performance.

6) Division of Training Epoch: To determine the optimal training epoch ratio between the SDP and TDP stages in the DSS data pipeline, we conducted experiments with various training ratios to avoid reliance on subjective empirical settings. As shown in Table IX, when only SDP is used (baseline), the model achieves 74.9% mAP@0.5:0.95, 93.1% mAP@0.5, and 82.8% mAP@0.75. Moreover, with the total training epochs fixed at 72, increasing the epochs for TDP to 4 improves the performance of DSSM to 76.3% mAP@0.5:0.95, 94.6% mAP@0.5, and 83.8% mAP@0.75. Furthermore, when the TDP epochs are further increased to 8, the model achieves gains of 1.9% in mAP@0.5:0.95, 1.9% in mAP@0.5, and 2.3% in mAP@0.75 compared to the baseline. However, when the TDP epochs are increased to 12, the model achieves 76.5% mAP@0.5:0.95, 94.7% mAP@0.5, and 84.0% mAP@0.75. Notably, training with TDP alone leads to difficulty in accurate convergence, achieving 72.7% mAP@0.5:0.95, 92.6% mAP@0.5, and 81.1% mAP@0.75, as prematurely introducing highly dynamic spatiotemporal flow variations and precise spatial geometric structures results in degraded performance.

D. Benchmark Comparisons

In this section, we first present comparative evaluations of our framework against contemporary state-of-the-art methods in Sections IV-D1–IV-D3 on the EMRS-Frame, EMRS-Event, and VisDrone-VID datasets, respectively. Notably, the comparison of VisDrone-VID is retrained and tested on the revised dataset to ensure a consistent evaluation environment and eliminate potential bias due to discrepancies in annotation noise. Subsequently, we analyze the comparative results of the data pipeline in the Section IV-D4. Additionally, we conduct experiments using identical training configurations as specified in Section IV-B for a fair comparison, with FPS evaluations performed on a single Nvidia GeForce RTX 3090 GPU.

1) Comparisons on the EMRS-Frame Dataset: As indicated in Table X, the widely used detectors YOLOv5 [30] and YOLOv8 [32] achieve 72.3% and 74.3% mAP@0.5:0.95, 91.1% and 92.9% mAP@0.5, and 73.9% and 72.0% mAP@0.75, respectively. Moreover, recently YOLO frameworks, including YOLOv11 [34] and YOLOv12 [35], achieve 75.8% and 75.0% mAP@0.5:0.95, 93.2% and 92.4% mAP@0.5, and 74.5% and 75.3% mAP@0.75. Furthermore, the advanced end-to-end detectors RT-DETR [39] and D-FINE [41] achieve 73.6% and 74.4% mAP@0.5:0.95, 92.2% and 93.6% mAP@0.5, and 74.5% and 74.7% mAP@0.75. However, these detection paradigms remain constrained to spatial information processing, lacking the capability to capture event streams with temporal priors. In contrast, our proposed model demonstrates superior performance across all complexity variants: Nano, Small, and Base configurations achieve 71.9%, 76.8%, and 77.6% mAP@0.5:0.95, 91.2%, 95.0%, and 96.5% mAP@0.5, alongside 77.0%, 85.1%, and 85.8% mAP@0.75, respectively, outperforming all contemporary state-of-the-art detectors. Notably, our approach achieves substantial improvements in both mAP@0.5 and mAP@0.75 metrics. The detection performance of our method is further visualized in Fig. 9 against other mainstream detectors, where the YOLO and DETR detectors suffer from false positives and false negatives issues, while DSSM achieves superior detection performance for challenging dark targets due to the capacity of streaming recursive inference.

2) Comparisons on the EMRS-Event Dataset: As indicated in Table XI, the widely used detectors YOLOv5 [30] and YOLOv8 [32] achieve 51.9% and 52.7% mAP@0.5:0.95, 69.2% and 69.5% mAP@0.5, and 54.1% and 53.7% mAP@0.75, respectively. Moreover, recently YOLO frameworks, including YOLOv11 [34] and YOLOv12 [35], achieve 55.0% and 54.4% mAP@0.5:0.95, 72.8% and 71.7% mAP@0.5, and 58.0% and 56.1% mAP@0.75. Furthermore, the advanced end-to-end detectors RT-DETR [39] and D-FINE [41] achieve 52.5% and 53.5% mAP@0.5:0.95, 71.1% and 72.0% mAP@0.5, and 55.2% and 56.2% mAP@0.75. However, these detection paradigms remain constrained to spatial information processing, lacking capability to capture event streams with temporal priors. In contrast, our

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:38:48 UTC from IEEE Xplore. Restrictions apply.

TABLE X

COMPARATIVE DETECTION PERFORMANCE ON THE EMRS-FRAME DATASET. THE BEST RESULT IS HIGHLIGHTED IN BOLD, AND THE SECOND BEST RESULT IS UNDERLINED

Model	Temporal	mAP @0.5:0.95	mAP @0.5	mAP @0.75	AP @S	AP @M	AP @L	FLOPs (G)	Params (M)	FPS
YOLOv6[Arxiv 2022]	✗	0.711	0.893	0.710	0.402	0.862	0.941	44.04	16.30	**181**
YOLOv5[Github 2020]	✗	0.723	0.911	0.739	0.453	0.878	0.958	23.84	9.11	147
DINO[ICLR 2023]	✗	0.728	0.920	0.727	0.426	0.843	0.949	119.67	45.21	32
RTDETR[CVPR 2024]	✗	0.736	0.922	0.745	0.438	0.866	0.957	30.00	46.19	55
YOLOv9[ECCV 2024]	✗	0.740	0.922	0.715	0.464	0.879	0.973	26.74	7.11	80
RTDETRv2[Arxiv 2024]	✗	0.741	0.933	0.738	0.447	0.889	0.954	30.00	46.19	56
YOLOv8[Github 2023]	✗	0.743	0.929	0.720	0.412	0.836	0.959	28.45	11.13	163
D-FINE[ICLR 2025]	✗	0.744	0.936	0.747	0.473	0.887	0.965	24.84	10.18	77
YOLOv10[NIPS 2024]	✗	0.748	0.926	0.752	0.442	0.896	0.959	24.46	8.04	128
YOLOv12[Arxiv 2025]	✗	0.750	0.924	0.753	0.447	0.874	0.980	19.27	9.08	84
YOLOv11[Arxiv 2024]	✗	0.758	0.932	0.745	0.452	0.860	0.960	21.31	9.41	128
DSSM-N (ours)	✓	0.719	0.912	0.770	0.423	0.856	0.943	**8.12**	**2.91**	143
DSSM-S (ours)	✓	0.768	0.950	0.851	0.465	0.891	0.967	28.34	10.74	126
DSSM-B (ours)	✓	**0.776**	**0.965**	**0.858**	**0.479**	**0.910**	**0.984**	77.35	23.70	118

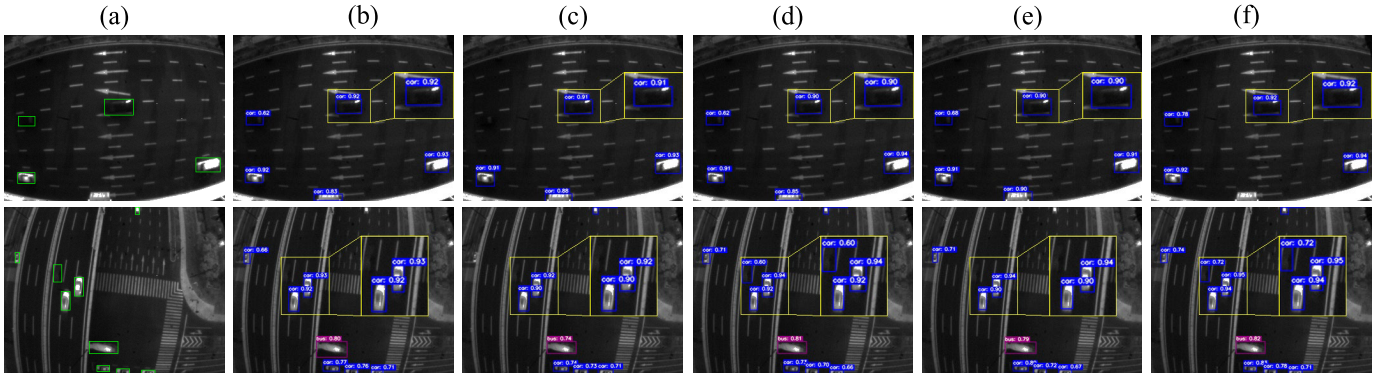


Fig. 9. Visualization of the benchmark comparison in the EMRS-Frame dataset. (a) Ground truth, (b) D-FINE, (c) YOLOv10, (d) YOLO11, (e) YOLO12, and (f) proposed model, DSSM.

proposed model demonstrates superior performance across all complexity variants: Nano, Small, and Base configurations achieve 50.9%, 55.1%, and 58.4% mAP@0.5:0.95, 69.0%, 73.3%, and 75.7% mAP@0.5, alongside 53.5%, 58.7%, and 62.4% mAP@0.75, respectively, outperforming all contemporary state-of-the-art detectors. Notably, our approach achieves substantial improvements in both mAP@0.5 and mAP@0.75 metrics. The detection performance of our method is further visualized in Fig. 10 against other mainstream detectors.

3) Comparisons on the VisDrone-VID Dataset: As delineated in Table XII, YOLOv5 [30] and YOLOv6 [31] achieve 10.2% and 11.5% mAP@0.5:0.95, and 20.7% and 23.1% mAP@0.5, alongside 9.4% and 9.9% mAP@0.75 with their lightweight architectures. Subsequent versions like YOLOv9 [42] and YOLOv10 [33] further improve the detection performance, reaching 11.0% and 10.5% mAP@0.5:0.95, 22.1% and 20.2% mAP@0.5, and 9.4% and 9.5% mAP@0.75. Moreover, recently YOLOv11 [34] and YOLOv12 [35] achieve 11.6% and 12.4% mAP@0.5:0.95, 23.6% and 24.6% mAP@0.5, and 9.8% and 10.8% mAP@0.75. Additionally, the end-to-end DETR-based detectors, including RT-DETR [39] and D-FINE [41], demonstrate competitive results, achieving 10.9% and 11.6% mAP@0.5:0.95, 21.8% and 24.4% mAP@0.5, and 9.6% and 9.9% mAP@0.75. Concurrently, our proposed model exhibits superior performance across its Nano, Small, and Base variants, achieving 11.3%, 13.3%, and 14.4% mAP@0.5:0.95, 23.4%, 26.7%, and 28.0% mAP@0.5, together with 9.2%, 11.9%, and 13.0% mAP@0.75, outperforming all contemporary state-of-the-art detectors. Comparative detection performance between our method and other advanced detectors is further visualized in Fig. 11.

4) Comparison of Data Pipeline: Currently, the mainstream object detection frameworks predominantly employ random sampling strategies during training. Although this approach partially mitigates overfitting risks to specific training set characteristics, it simultaneously disrupts inherent temporal dependencies within sequential data and undermines the capacity of the model to extract meaningful temporal embeddings from consecutive frames. Conversely, sequential sampling methodology preserves temporal continuity and contextual semantics by processing data in chronological order, while this sequential nature introduces dependency toward specific data distributions, consequently constraining model generalization performance. Furthermore, a random batch processing strategy effectively alleviates the bias inherent in sequential sampling by extracting frames from identical

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:38:48 UTC from IEEE Xplore. Restrictions apply.

TABLE XI

COMPARATIVE DETECTION PERFORMANCE ON THE EMRS-EVENT DATASET. THE BEST RESULT IS HIGHLIGHTED IN BOLD, AND THE SECOND BEST RESULT IS UNDERLINED

Model	Temporal	mAP @0.5:0.95	mAP @0.5	mAP @0.75	AP @S	AP @M	AP @L	FLOPs (G)	Params (M)	FPS
YOLOv9[ECCV 2024]	✗	0.515	0.687	0.525	0.263	0.696	0.919	26.74	7.17	82
ASTMNet[TIP 2022]	✓	0.516	0.690	0.521	0.207	0.735	0.923	158.57	107.35	27
DINO[ICLR 2023]	✗	0.519	0.707	0.548	0.226	0.735	0.923	119.67	45.21	31
YOLOv5[Github 2020]	✗	0.519	0.692	0.541	0.227	0.744	0.916	23.84	9.11	158
RTDETR[CVPR 2024]	✗	0.525	0.711	0.552	0.230	0.724	0.938	30.00	46.19	54
YOLOv8[Github 2023]	✗	0.527	0.695	0.537	0.237	0.713	0.962	28.45	11.13	167
YOLOv6[Arxiv 2022]	✗	0.528	0.706	0.548	0.244	0.736	0.963	44.04	16.30	**178**
RTDETRv2[Arxiv 2024]	✗	0.528	0.704	0.552	0.235	0.740	0.941	30.00	46.19	55
YOLOv10[NIPS 2024]	✗	0.533	0.705	0.546	0.225	0.736	0.965	24.46	8.04	131
D-FINE[ICLR 2025]	✗	0.535	0.720	0.562	0.245	0.757	0.949	24.84	10.18	79
YOLOv12[Arxiv 2025]	✗	0.544	0.717	0.561	0.243	0.781	0.943	19.27	9.08	85
YOLOv11[Arxiv 2024]	✗	0.550	0.728	0.580	0.258	0.795	0.952	21.31	9.41	133
DSSM-N (ours)	✓	0.509	0.690	0.535	0.207	0.715	0.899	**8.12**	**2.91**	155
DSSM-S (ours)	✓	0.551	0.733	0.587	0.261	0.797	0.950	28.34	10.74	138
DSSM-B (ours)	✓	**0.584**	**0.757**	**0.624**	**0.284**	**0.822**	**0.968**	77.35	23.70	123

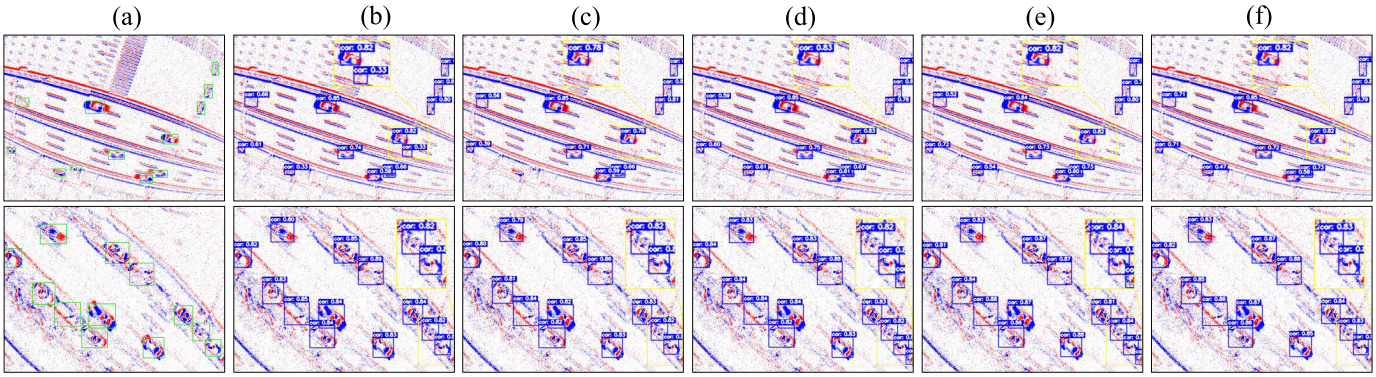


Fig. 10. Visualization of the benchmark comparison in the EMRS-Event dataset. (a) Ground truth, (b) D-FINE, (c) YOLOv10, (d) YOLO11, (e) YOLOv12, and (f) proposed model, DSSM.

temporal positions across different video sequences, yet brings challenges for efficient multithreaded parallel data loading.

To address these limitations of existing approaches, our proposed DSS architecture synthesizes the complementary advantages of randomized batch processing and sequential sampling, maintaining temporal modeling capabilities while supporting high-concurrence multithreaded data-loading mechanisms. Meanwhile, despite employing a dual-phase training strategy, the architecture does not introduce significant additional memory overhead during training thanks to the dual-phase data pipeline operating as an online strategy switch within the same data-loading framework, rather than preserving two independent parallel data-processing streams. In other words, a single data loader instance is retained in memory throughout the training process, with internal sampling methods and augmentation modules dynamically adjusted according to the training progression. This design effectively avoids redundant memory occupation caused by maintaining multiple data pipelines. Furthermore, the multithreaded data prefetching mechanism adopted in this architecture efficiently prepares the next batch of data in the background while GPU computation is ongoing, achieving a high degree of overlap between data loading and model training. Therefore, the emphasis of the DSS design is on the temporal scheduling of training strategies and the dynamic switching of augmentation methods with memory consumption showing no significant difference compared to conventional single-phase pipelines.

Specifically, as shown in Table XIII, sequential sampling achieves 65.2% and 45.8% mAP@0.5:0.95 on the frame and event datasets of EMRS, respectively, with a resource usage of 9.0-GB CPU and 12.22-GB GPU memory, while random sampling strategy improve by 5.9% and 3.4% mAP@0.5:0.95 on the corresponding datasets with a resource usage of 7.2-GB CPU and 13.82-GB GPU memory. Furthermore, the random batch strategy achieves 73.6% and 52.1% mAP@0.5:0.95 on the frame and event dataset of EMRS with a resource usage of 2.9-GB CPU and 13.80-GB GPU memory. Notably, Fig. 12 illustrates the convergence trajectories of detection performance during training for four distinct methodologies, with the DSS strategy exhibiting superior convergence efficiency. Furthermore, the DSS architecture demonstrates substantial performance improvement during the final epochs, attributable to its embedded spatiotemporal joint training mechanism within the data pipeline. Simply, this mechanism employs

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:38:48 UTC from IEEE Xplore. Restrictions apply.

TABLE XII

COMPARATIVE DETECTION PERFORMANCE ON THE VISDRONE-VID DATASET. THE BEST RESULT IS HIGHLIGHTED IN BOLD, AND THE SECOND BEST RESULT IS UNDERLINED

Model	Temporal	mAP @0.5:0.95	mAP @0.5	mAP @0.75	AP @S	AP @M	AP @L	FLOPs (G)	Params (M)	FPS
YOLOv5[Github 2020]	✗	0.102	0.207	0.094	0.032	0.089	0.186	55.19	46.19	75
YOLOv10[NIPS 2024]	✗	0.105	0.202	0.095	0.039	0.094	0.188	57.00	8.04	89
RTDETR[CVPR 2024]	✗	0.109	0.218	0.096	0.043	0.110	0.187	285.26	46.19	18
YOLOv9[ECCV 2024]	✗	0.110	0.221	0.094	0.057	0.120	0.175	61.87	7.11	62
YOLOv6[Arxiv 2022]	✗	0.115	0.231	0.099	0.051	0.124	0.193	102.80	9.41	85
YOLOv11[Arxiv 2024]	✗	0.116	0.236	0.098	0.054	0.132	0.203	46.27	46.19	91
D-FINE[ICLR 2025]	✗	0.116	0.244	0.099	0.049	0.145	0.201	91.17	10.18	46
YOLOv8[Github 2023]	✗	0.118	0.241	0.101	0.055	0.135	0.186	65.79	11.13	**93**
YOLOv12[Arxiv 2025]	✗	0.124	0.246	0.108	0.040	0.144	0.229	44.70	9.08	60
DSSM-N (ours)	✓	0.113	0.234	0.092	0.050	0.132	0.206	**18.67**	**2.91**	81
DSSM-S (ours)	✓	0.133	0.267	0.119	0.065	0.169	0.236	65.10	10.74	74
DSSM-B (ours)	✓	**0.144**	**0.280**	**0.130**	**0.078**	**0.184**	**0.253**	177.91	23.70	56

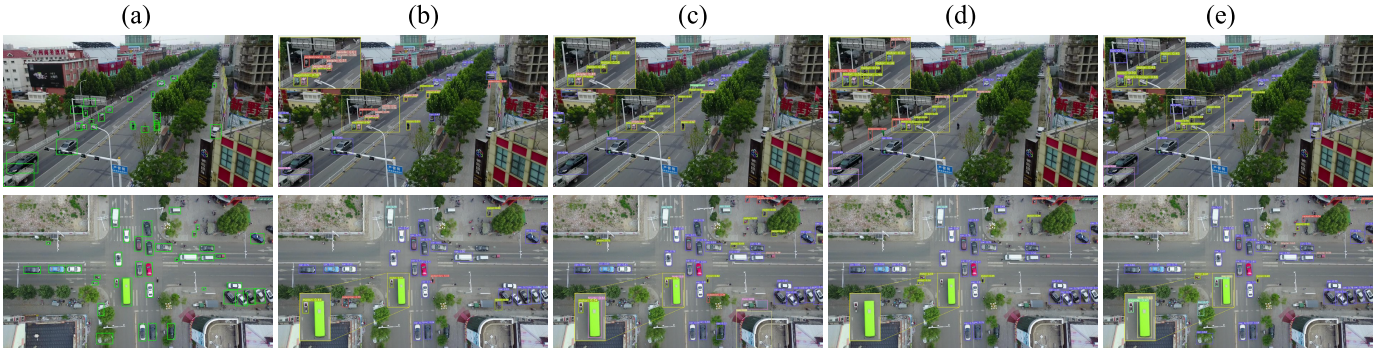


Fig. 11. Visualization of the benchmark comparison in the VisDrone-VID dataset. (a) Ground truth, (b) D-FINE, (c) YOLO11, (d) YOLOv12, and (e) proposed model, DSSM.

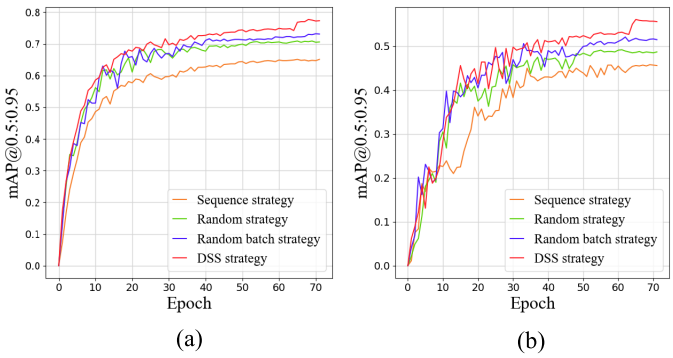


Fig. 12. Comparison of the training accuracy. (a) Comparison results of mAP@0.5:0.95 on the EMRS-frame dataset and (b) comparison results of mAP@0.5:0.95 on the EMRS-event dataset, and the DSS strategy manifests accelerated convergence properties and superior detection performance.

progressive feature enhancement to achieve decoupled two-phase training for spatiotemporal modeling, thereby effectively overcoming the representational learning bottleneck inherent in conventional detection methods caused by spatial regression tasks.

V. DISCUSSION

In this study, we propose a dynamic streaming architecture designed to address remote sensing visual perception challenges arise from viewpoint variations and complex backgrounds. By introducing the dual-stage streaming data pipeline, the selective memory module, and the composite spatiotemporal loss function, our approach achieves remarkable detection performance. However, these components also warrant discussion to analyze their potential limitations.

TABLE XIII

COMPARISON OF THE DSS DATA PIPELINE ON THE EMRS DATASET. THE BEST DETECTION PERFORMANCE IN BOLD

Data pipeline	Temporal	CPU (GB)	GPU (GB)	mAP@0.5:0.95	
				Frame	Event
Sequence strategy	✗	9.0	12.22	0.652	0.458
Random strategy	✗	7.2	13.82	0.711	0.492
Random batch strategy	✗	**2.9**	13.80	0.736	0.521
DSS strategy (ours)	✓	5.8	**10.73**	**0.768**	**0.551**

The constructed DSS data pipeline significantly enhances the capacity of the model for fine-grained localization of targets through a phased spatiotemporal state-awareness mechanism. Compared to conventional random sampling pipelines, the proposed pipeline utilizes spatial stream to accelerate the convergence of static representations, while applying temporal stream to inject dynamic spatiotemporal details, thereby enabling more stable feature extraction in complex scenarios. To quantify this advantage, we conduct a systematic comparison of the two pipelines using confusion matrix analysis.

As illustrated in Fig. 13, the DSS data pipeline demonstrates outstanding performance in reducing both false positives and false negatives. Specifically, the confusion matrix reveals that

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:38:48 UTC from IEEE Xplore. Restrictions apply.

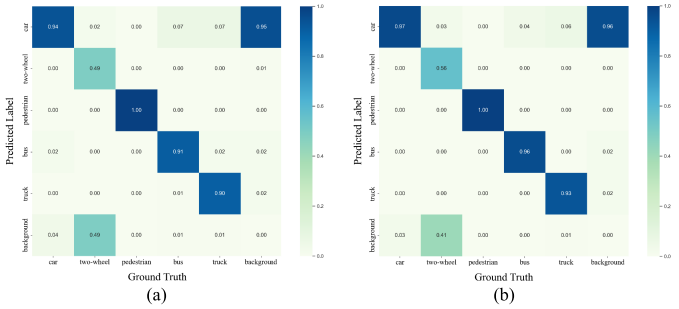


Fig. 13. Visualization of the confusion matrix. (a) and (b) Confusion matrices of random and DSS strategies. Moreover, the rows correspond to GT labels, while columns represent predicted categories.

the random sampling pipeline tends to introduce spatiotemporal inconsistencies due to its irrelevant sequential dependency, leading to increased interclass confusion. In contrast, the DSS data pipeline effectively reduces the sensitivity of the model to specific data sequences via batch parallelization, thereby improving both generalization capability and convergence speed. In other words, this improvement underscores the critical role of the DSS data pipeline in streaming object detection since it provides richer contextual priors and improves detection robustness through the synergistic optimization of spatiotemporal states.

Furthermore, the incorporation of the SMM and the composite spatiotemporal loss function further strengthens the ability of the model to capture long-range temporal dependencies. Specifically, the SMM adaptively encodes spatiotemporal cues through decoupled gating units, while the spatiotemporal loss function guides the model toward stable learning dynamics in complex streams via state-smoothing constraints, motion-response constraints, and gating-conflict penalties. These components synergistically with the DSS data pipeline to facilitate fine-grained representation learning. Nevertheless, the optimal weighting of each component in the spatiotemporal loss is closely dependent on the characteristics of specific scenarios, where factors such as dynamic object density and background complexity may lead to variations in weight assignment. Therefore, this dependency necessitates meticulous fine-tuning in practical applications to prevent overfitting or underfitting.

Eventually, this study offers valuable insights into streaming object detection research through innovative architectural design and comprehensive dataset refinement. However, there are still some limitations. The primary restriction lies in the single-scale SMM module, which is unable to fully leverage the rich positional and detailed cues in shallow features and fails to establish independent temporal memory flows across multiple scales. This may constrain the ability of the model to capture finer spatiotemporal contexts when capturing fast-moving or intermittently small objects. Therefore, our future work will investigate interaction mechanisms for cross-scale memory states to improve synergy and complementarity of spatiotemporal information across different resolutions, while exploring adaptive weight adjustment mechanisms to improve the cross-scenario generalization capability of the model.

VI. Conclusion

In aerial remote sensing scenarios for UAVs, abrupt viewpoint changes, background clutter, occlusions, and motion blur bring severe challenges to visual perception. To address these challenges, we introduce a novel detection paradigm based on the dynamic streaming architecture, which integrates three principal components: 1) dual-stage streaming data pipeline (DSS data pipeline) that models static representations and dynamic spatiotemporal evolution via two pathways to achieve fine-grained state awareness and efficient feature refinement; 2) SMM that adaptively fuses critical spatiotemporal context from historical frames through a decoupled gating mechanism to capture long-range temporal dependencies; and 3) composite spatiotemporal loss function that jointly enforces state-smoothing constraint, motion-response constraint, and gating-conflict penalty to steer localization and discrimination under complex dynamic conditions. Moreover, we perform a systematic reannotation of the VisDrone-VID dataset to improve evaluation fidelity. Experimental results on the EMRS and VisDrone-VID benchmarks demonstrate that the proposed DSSM outperforms all state-of-the-art methods, corroborating the effectiveness and generalizability as a streaming detection framework for challenging UAV remote sensing environments.

References

[1] S. Jing et al., "ESVT: Event-based streaming vision transformer for challenging object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5607113.

[2] K. Liu, T. Li, and D. Peng, "Aerial image object detection based on RGB-infrared multibranch progressive fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5402714.

[3] H. Liu, D. He, Y. Zhao, Q. Shi, and X. Liu, "PSODNet: Pretrained scene-aware object detection for optical remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4707615.

[4] Z. Wei et al., "Similar category enhancement network for discrimination on small object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5651718.

[5] D. Zhang, N. Li, M. Zhao, and K. Xu, "Improved yolov8 algorithm was used to segment cucumber seedlings under complex artificial light conditions," *IEEE Access*, vol. 13, pp. 83944–83955, 2025.

[6] R. Kou et al., "Infrared small target segmentation networks: A survey," *Pattern Recognit.*, vol. 143, 2023, Art. no. 109788.

[7] L. Liu, C. Chu, C. Chen, and S. Huang, "Marineyolo: Innovative deep learning method for small target detection in underwater environments," *Alexandria Eng. J.*, vol. 104, pp. 423–433, 2024.

[8] X. Yin and L. Chen, "Image object detection method based on improved faster R-CNN," *J. Circuits, Syst. Comput.*, vol. 33, no. 7, 2024, Art. no. 2450130.

[9] J. Han, Y. Yu, K. Liang, and H. Zhang, "Infrared small-target detection under complex background based on subblock-level ratio-difference joint local contrast measure," *Opt. Eng.*, vol. 57, p. 1, Oct. 2018.

[10] H. Fang et al., "Infrared small uav target detection based on depthwise separable residual dense network and multiscale feature fusion," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–20, 2022.

[11] J. Pan, Y. Gu, G. Gao, and S. Wu, "Real-time vehicle detection in satellite videos: Transitioning from large scenes to clusters," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5628813.

[12] X. Xu et al., "EMTrack: Event-guide multimodal transformer for challenging single-object tracking," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5626914.

[13] S. Jing et al., "Hyper lightweight neural networks towards spike-driven deep residual learning," *Knowledge-Based Syst.*, vol. 327, Oct. 2025, Art. no. 114099.

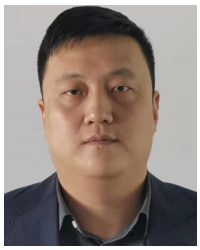
[14] M. Dang, G. Liu, H. Li, D. Wang, R. Pan, and Q. Wang, "PRA-Det: Anchor-free oriented object detection with polar radius representation," *IEEE Trans. Multimedia*, vol. 27, pp. 145–157, 2025.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:38:48 UTC from IEEE Xplore. Restrictions apply.

[15] M. Cannici, M. Ciccone, A. Romanoni, and M. Matteucci, "A differentiable recurrent surface for asynchronous event-based data," 2020, *arXiv:2001.03455*.
[16] Y. Shao et al., "EventVAD: Training-free event-aware video anomaly detection," in *Proc. 33rd ACM Int. Conf. Multimedia*, New York, NY, USA, Oct. 2025, pp. 2586–2595, doi: 10.1145/3746027.3754500.
[17] D. Li, Y. Tian, and J. Li, "SODFormer: Streaming object detection with transformer using events and frames," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 11, pp. 14020–14037, Nov. 2023.
[18] J. Li, J. Li, L. Zhu, X. Xiang, T. Huang, and Y. Tian, "Asynchronous spatio-temporal memory network for continuous event-based object detection," *IEEE Trans. Image Process.*, vol. 31, pp. 2975–2987, 2022.
[19] Y. Zheng, B. Zhong, Q. Liang, Z. Mo, S. Zhang, and X. Li, "ODTrack: Online dense temporal token learning for visual tracking," in *Proc. AAAI Conf. Artif. Intell.*, 2024, vol. 38, no. 7, pp. 7588–7596.
[20] M. Gehrig and D. Scaramuzza, "Recurrent vision transformers for object detection with event cameras," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 13884–13893.
[21] R. Zhang et al., "S3OD: Size-unbiased semi-supervised object detection in aerial images," *ISPRS J. Photogramm. Remote Sens.*, vol. 221, pp. 179–192, Mar. 2025.
[22] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997.
[23] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 1–18.
[24] D. Du et al., "VisDrone-DET2019: The vision meets drone object detection in image challenge results," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 213–226.
[25] N. Zubic, M. Gehrig, and D. Scaramuzza, "State space models for event cameras," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 5819–5828.
[26] J. Chung, C. Gulcehre, K. Cho, and Y. Bengio, "Empirical evaluation of gated recurrent neural networks on sequence modeling," 2014, *arXiv:1412.3555*.
[27] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.
[28] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.
[29] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.*Cham, Switzerland: Springer, 2020, pp. 213–229.
[30] R. Khanam and M. Hussain, "What is yolov5: A deep look into the internal features of the popular object detector," 2024, *arXiv:2407.20892*.
[31] C. Li et al., "YOLOv6: A single-stage object detection framework for industrial applications," 2022, *arXiv:2209.02976*.
[32] G. Jocher, A. Chaurasia, and J. Qiu. (2023). *Ultralytics YOLOv8*. [Online]. Available: https://github.com/ultralytics/ultralytics
[33] H. Chen et al., "YOLOv10: Real-time end-to-end object detection," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 107984–108011.
[34] R. Khanam and M. Hussain, "YOLOv11: An overview of the key architectural enhancements," 2024, *arXiv:2410.17725*.
[35] Y. Tian, Q. Ye, and D. Doermann, "YOLOv12: Attention-centric real-time object detectors," 2025, *arXiv:2502.12524*.
[36] Z. Luo, Z. Fu, Z. Huang, W. Fu, Z. Zhu, and X. Chen, "A day-night cross-modal network for robust commodity recognition under low-light illumination," *Eng. Appl. Artif. Intell.*, vol. 164, Jan. 2026, Art. no. 113164. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0952197625031951
[37] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," 2020, *arXiv:2010.04159*.
[38] H. Zhang et al., "DINO: DETR with improved DeNoising anchor boxes for end-to-end object detection," 2022, *arXiv:2203.03605*.
[39] Y. Zhao et al., "DETRs beat YOLOs on real-time object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 16965–16974.
[40] W. Lv, Y. Zhao, Q. Chang, K. Huang, G. Wang, and Y. Liu, "RT-DETRv2: Improved baseline with bag-of-freebies for real-time detection transformer," 2024, *arXiv:2407.17140*.
[41] Y. Peng, H. Li, P. Wu, Y. Zhang, X. Sun, and F. Wu, "D-Fine: Redefine regression task of detrs as fine-grained distribution refinement," 2025, *arXiv:2410.13842*.
[42] C.-Y. Wang, I.-H. Yeh, and H.-Y. M. Liao, "YOLOv9: Learning what you want to learn using programmable gradient information," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 1–21.

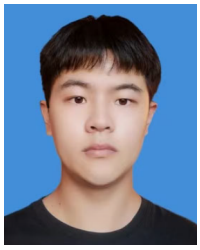
Shilong Jing is currently pursuing the Ph.D. degree with Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences, Changchun, China.

His research interests include object detection and recognition, dynamic vision sensors, and image semantic segmentation.



Yuchen Zhao received the Ph.D. degree from Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences, Changchun, China, in 2016.

He joined Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences, in 2016. He is currently the Deputy Director of the Imaging Electronics Laboratory, an Associate Researcher, and a Master's Supervisor of Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences. His research interests include space optics, optical imaging, and dynamic vision sensors.



Zhuoxian Li received the B.E. degree in mechanical and electronic engineering from North Minzu University, Yinchuan, China, in 2024. He is currently pursuing the master's degree with Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences, Changchun, China.

His research interests are primarily focused on neural network algorithms and basic vision such as image segmentation and object detection.

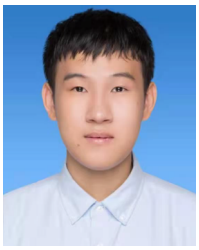
Xiangzhi Li received the Ph.D. degree from Harbin Institute of Technology, Harbin, China, in 2010.

He joined Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences, Changchun, China, in 2010. He is currently an Associate Research Fellow and a Master's Supervisor with the Imaging Electronics Laboratory, Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences. His research interests include space camera overall design technology and image processing technology.

Xianda Xu received the B.E. degree in optoelectronic information science and engineering from Changchun University of Science and Technology, Changchun, China, in 2023. He is currently pursuing the master's degree with Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences, Changchun.

His research interests are primarily focused on neural network algorithms and basic vision such as deblurring and image reconstruction.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:38:48 UTC from IEEE Xplore. Restrictions apply.



Hechong Wang received the B.E. degree from the School of Information and Computing Science, Dalian University of Technology, Dalian, China, in 2022. He is currently pursuing the Ph.D. degree with Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences, Changchun, China.

His research interests include computer vision and image processing.

Hengyi Lv received the B.E. degree in automation from Dalian University of Technology, Dalian, China, in 2007, the M.S. degree in automatic control from Harbin Institute of Technology, Harbin, China, in 2009, and the Ph.D. degree in mechatronic engineering from the University of Chinese Academy of Sciences, Beijing, China, in 2018.

He is currently the Director of the Imaging Electronics Laboratory, a Researcher, and a Doctoral Supervisor of the Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences, Changchun, China. His research interests include electronics of space cameras, space remote sensing imaging technology, dynamic vision sensor applications, and artificial intelligence (AI) advanced imaging technology.

Ming Sun received the Ph.D. degree from Jilin University, Changchun, China, in 2023.

His research interests include space remote sensing imaging technology, machine vision, and pattern recognition.

Yisa Zhang received the Ph.D. degree from Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences, Beijing, China, in 2023.

He is currently an Assistant Researcher with Changchun Institute of Optics, Fine Mechanics and Physics. His research interests are remote sensing image processing and asynchronous spatiotemporal data processing based on event cameras.

Yang Feng received the B.E. degree from Northeast University, Shenyang, China, in 2016, and the Ph.D. degree from the University of Chinese Academy of Sciences, Beijing, China, in 2021.

He is currently an Assistant Researcher with Changchun Institute of Optics, Fine Mechanics and Physics, Changchun, China. His research interests include sensor circuit design and dynamic visual sensing imaging systems.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:38:48 UTC from IEEE Xplore. Restrictions apply.